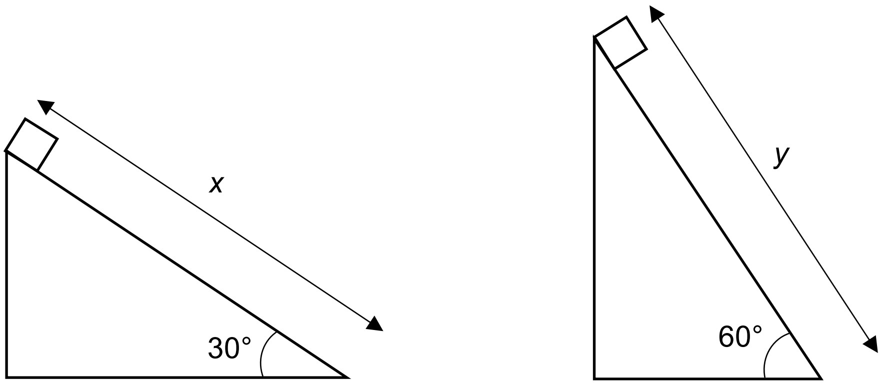
D m^{-1} s^{-3} kg

3 Two identical blocks are released from rest from the top of two ramps as shown below.



Ignoring friction, what must be the ratio of y/x so that both blocks arrive at the end of the ramps at the same time?

A $\frac{1}{\sqrt{3}}$ **B** $\sqrt{2}$ **C** $\sqrt{3}$ **D** 3

Ans: C

$s_y = u_y t + \frac{1}{2} a_y t^2$

$s_y = \frac{1}{2} a_y t^2$ since u = 0

$t = \sqrt{(2s_y / a_y)}$

$\sqrt{[2 (x/\sin 30°) / a_y]} = \sqrt{[2 (y/\sin 60°)/ a_y]}$

$y/x = \sin 60° / \sin 30° = \sqrt{3}$